A Selected proof routes

Target, law, loss, feasible family
Original units and M

Weak-error route
Equality counts as wrong
$R^s = [A^s + \tau(M - N)]/M$

Equality-neutral route
Strict error + distinct withholding
$R^s_{\mathrm{TA}} = [A^s_{\mathrm{TA}} + \tau(T - Q)]/M$

Own-risk optimum
Eligible competitors
Same law and loss
Same family

Count hull
(N, A)
Fixed family
Whole reports

Guarded minimizer exists
Full at-most G
No extras
Retain pairs
Existence only

Distinct-mass hull (Q, A)
Fixed family
Whole reports

Extra for half cost:
complementarity; full $\mathcal{A}_n$
all strict children feasible

Extra for half cost:
full-profile class domain
all class-strict children feasible
not guard existence alone

Selected prerequisites only; finite-sum risk needs no criterion independence.

B Why every required refinement is optimal

One law, one loss and one family
g^* is an EXACT half-cost minimizer.
EVERY required child is feasible.

WEAK: full unit refinements
$\bar{e}_{ij} + \bar{e}_{ji} = 1$
Full unrestricted $\mathcal{A}_n$
All strict unit-order children
$q = M - N(g^*)$

NEUTRAL: full class refinements
$c_{ij} + c_{ji} = \delta_{ij}$
Full active-profile equality classes
All class-strict children feasible
Original unit-pair multiplicities
$q = T - Q(g^*)$

Uniform average over ALL required children
Mean added error $= q/2$
Half-cost withholding reduction $= q/2$
Mean $R(r) = R(g^*) = R^*$

Exact optimum + child feasibility
$R(r) - R^* \geq 0$ for every child

Uniform mean of all child gaps
$\frac{1}{L}\sum_{\ell=1}^{L} [R(r_\ell) - R^*] = 0$

Every required child: $R(r) = R^*$

Endpoints: weak $N = M$; neutral $Q = T$ (raw N may stay below M).
Guarded $T = 0$: one class/all-withhold, point frontier; Q/T undefined.

Uniform averaging is a proof device, not a reporting policy. Caps, fixed orders, protections and T11 need separate arguments. No arbitrary-refinement guarantee for an approximate optimizer.